A Latent Profile Analysis of Attachment Dimensions and Their Associations with Sleep, Mood, and Fatigue

Iris Chen

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Abstract

Adult attachment has been linked to sleep quality, mood, and health-related behaviors, but much of the existing research examines attachment anxiety and attachment avoidance as separate predictors. The present study used a person-centered approach to examine whether distinct attachment profiles were associated with differences in sleep, mood, fatigue, and daily functioning. Secondary data were drawn from an online study of adults in romantic relationships. Latent profile analysis was conducted using 12 attachment indicators from the Experiences in Close Relationships Questionnaire. A four-profile solution was selected and interpreted as Dismissive-leaning, Fearful-avoidant, Anxious-preoccupied, and Secure. The profiles differed across several sleep and mood outcomes. The Fearful-avoidant and Anxious-preoccupied profiles generally reported more sleep problems, poorer sleep hygiene, and higher anxious and depressed mood than the Secure profile. The Fearful-avoidant profile showed the most consistently negative pattern across outcomes, including poorer sleep efficiency, greater mood symptoms, and more difficulty with enthusiasm and daily functioning. In contrast, the Secure profile generally showed the most adaptive pattern, including fewer sleep problems, better mood outcomes, and lower fatigue. These findings suggest that attachment anxiety and avoidance may combine into meaningful profiles that are differentially associated with sleep and psychological well-being. A person-centered approach may therefore provide a more nuanced understanding of how adult attachment patterns relate to sleep, mood, and fatigue.

1 Introduction

Sleep quality is closely related to both physical and psychological well-being. Poor sleep quality and sleep disturbances have been consistently linked to a range of mental health problems, including depression, anxiety, and stress (Hertenstein et al., 2019). Sleep difficulties are also related to daytime functioning and fatigue, making sleep an important outcome to examine in relation to broader psychological and relational factors (Lavidor et al., 2003). Although sleep is often studied as a biological or behavioral process, growing evidence suggests that close relationships and individual differences in relationship functioning may also shape sleep health.

Close relationships provide an important context for understanding sleep and well-being. Social connection and disconnection can affect health through emotional, behavioral, and physiological pathways, including stress regulation, health behaviors, and biological stress responses (Pietromonaco & Collins, 2017). Similar patterns have been observed for sleep, as the quality of close relationships and romantic relationships has been linked to sleep quality through mechanisms such as felt security, emotional responses, self-regulation, and biological processes (Kent et al., 2015; Xie & Feeney, 2024). Because sleep may be influenced by feelings of safety, stress, and interpersonal security, attachment theory offers a useful framework for understanding individual differences in sleep, mood, and fatigue.

Attachment theory was originally developed to explain early caregiver–child bonds, but it was later extended to adult close relationships (Bowlby, 1985; Hazan & Shaver, 1987). In adulthood, attachment is commonly conceptualized along two relatively independent dimensions: attachment anxiety and attachment avoidance (Fraley et al., 2015). Attachment anxiety reflects fears of rejection, abandonment, and not being cared for, whereas attachment avoidance reflects discomfort with closeness, dependence, and intimacy. These dimensions reflect internal working models of the self and others that guide emotional regulation, coping strategies, and interpersonal behavior (Adams et al., 2014; Mikulincer & Shaver, 2019). Low levels of both anxiety and avoidance are generally associated with secure attachment, while different combinations of the two dimensions can form theoretically meaningful attachment styles, including secure, dismissing, preoccupied, and fearful attachment (Bartholomew & Horowitz, 1991).

Attachment processes are theorized to play a key role in sleep by influencing emotional regulation and physiological arousal at bedtime. Secure attachment, characterized by feelings of safety and trust, is thought to support lower arousal levels that facilitate restful sleep, whereas insecure attachment is often associated with heightened vigilance and emotional dysregulation that can interfere with sleep (Troxel, 2010). Individuals high in attachment anxiety may rely on hyperactivating strategies, meaning they may become more worried, emotionally activated, and reassurance-seeking during relationship stress. This pattern may contribute to elevated pre-sleep arousal and greater difficulty initiating and maintaining sleep (Rogojanski et al., 2013). In contrast, individuals high in attachment avoidance often rely on deactivating strategies that suppress emotional needs and distance the self from others. Although anxiety and avoidance may operate through different coping strategies, both forms of insecure attachment have been linked to stress reactivity and poorer sleep outcomes (Adams et al., 2014; Hu et al., 2024).

Empirical evidence supports the link between attachment insecurity, sleep, and psychological functioning. Prior research has found that insecure attachment is associated with poorer sleep quality and sleep disturbances across adulthood (Adams et al., 2014; Hu et al., 2024). Attachment insecurity has also been linked to depressive and anxious symptoms, suggesting that attachment-related emotion regulation patterns may be relevant not only to sleep but also to mood

(Jinyao et al., 2012; Mikulincer & Shaver, 2019). In addition, attachment may influence health-related behaviors, including behaviors that support or interfere with sleep. For example, insecure attachment has been associated with riskier health behaviors and fewer preventive health behaviors, while secure attachment is generally linked to more adaptive coping and health behavior patterns (Huntsinger & Luecken, 2004; Pietromonaco & Beck, 2019). This suggests that attachment may be relevant to sleep both directly, through emotional arousal, and indirectly, through sleep-related habits such as sleep hygiene.

However, findings in adult attachment and sleep research have not always been consistent. One possible reason is that many studies use variable-centered approaches, examining attachment anxiety and attachment avoidance as separate predictors. This approach is useful, but it may miss the fact that individuals can show different combinations of anxiety and avoidance. For example, one person may be high in anxiety but low in avoidance, while another may be high in both anxiety and avoidance. Both individuals may be considered insecurely attached, but their emotional, behavioral, and sleep-related outcomes may differ. Therefore, a person-centered approach may provide a more nuanced understanding of how attachment relates to sleep, mood, and fatigue.

Latent profile analysis is one person-centered method that can identify unobserved subgroups of individuals based on patterns across multiple continuous indicators (Spurk et al., 2020). In the context of adult attachment, latent profile analysis allows attachment anxiety and avoidance indicators to be examined simultaneously, rather than assuming that each dimension operates in the same way for all individuals. This is especially appropriate because attachment theory already suggests that anxiety and avoidance form a two-dimensional space that can give rise to distinct attachment patterns, such as secure, dismissing, preoccupied, and fearful attachment (Bartholomew & Horowitz, 1991; Mikulincer & Shaver, 2016). Applying this framework to sleep hygiene, sleep quality, mood, and fatigue may offer new insights into how attachment-based profiles correspond to different patterns of psychological and sleep-related functioning.

The present study builds on prior research by using latent profile analysis to identify attachment-based profiles from the 12 attachment indicators. It then examines whether these profiles differ in sleep quality, sleep problems, sleep efficiency, sleep hygiene, mood symptoms, fatigue, and daily functioning. By taking a person-centered approach, this study aims to provide a more nuanced understanding of how different combinations of attachment anxiety and avoidance are associated with sleep, mood, and fatigue outcomes.

2 Method

2.1 Participants and procedures

The present study used secondary data from an OSF-hosted project on adult attachment, sleep hygiene, and sleep quality. The dataset was originally collected for a cross-sectional study examining whether sleep hygiene mediated the association between adult attachment dimensions and sleep quality. The original study recruited participants through Prolific, and participants completed online questionnaires through Qualtrics. Eligible participants were adults over the age of 18 who were currently in an intimate relationship for at least three months and were able to speak English. The final sample included 220 participants. The mean age of participants was 32.35 years old (SD = 8.91), with ages ranging from 18 to 62. Slightly over half of the sample was female, and most participants reported being in a relationship for more than three years.

Table 1: Demographic characteristics of the sample

Category	**N = 220**[1]
age_group	
18–29	44%
30–39	36%
40–49	16%
50–59	3.6%
60–69	0.9%
70–79	0%
80–89	0%
gender	
Female	55%
Male	45%
ethnicity	
Asian	5.5%
Black, Black British, Caribbean, or African	42%
From multiple races	3.2%
Other	0.9%
White	49%
relationship_duration	
More than 3 years	67%
One to three years	26%
Two months to a year	6.8%
live	
no	20%
Other	0.9%
yes	79%

[1] Percentages are shown.

2.2 Measures

Attachment dimensions

Adult attachment was measured using the 12-item Experiences in Close Relationships Questionnaire (ECR-12). This measure includes two attachment dimensions: attachment anxiety and attachment avoidance. Items were rated on a 7-point scale, with higher scores indicating higher levels of attachment anxiety or avoidance. In the original study, internal consistency was acceptable for both the anxious attachment dimension and the avoidant attachment dimension.

Sleep quality
Sleep quality was measured using the shortened Pittsburgh Sleep Quality Index. This measure includes questions about sleep duration, sleep onset, sleep quality, sleep problems, daytime functioning, and sleep efficiency. Higher total scores indicate poorer sleep quality.

Mood
Mood symptoms were measured using the Patient Health Questionnaire-4 (PHQ-4), a brief measure of anxiety and depressive symptoms. Higher scores indicate greater mood symptoms.

Fatigue
Fatigue was measured using a single item from the Multidimensional Mood Questionnaire. Participants rated whether they generally felt exhausted or full of energy over the past two weeks. In this study, this variable was used as an indicator of fatigue or energy level.

2.3 Data Preparation

For the present analysis, the 12 ECR attachment items were used as indicators in the latent profile analysis. These items included both attachment anxiety and attachment avoidance items. Before estimating the profiles, missing data were handled using single imputation. The attachment items were then standardized so that each item had a mean of 0 and a standard deviation of 1. Standardization was used because the latent profile model compares patterns across items, and putting the items on the same scale makes the profile differences easier to interpret.

2.4 Analytic Strategy

A latent profile analysis (LPA) was conducted in R using the tidyLPA package to explore heterogeneity in adult attachment patterns using the 12 items from the Experiences in Close Relationships Questionnaire (ECR-12). These indicators reflected both attachment anxiety and attachment avoidance. Before conducting the LPA, missing data were addressed using single imputation, and all attachment indicators were standardized. Standardization allowed the profiles to be interpreted as relative elevations or reductions across attachment-related items.

Models with one to six profiles were estimated. The following indices were used to evaluate model fit and guide profile selection: (1) Akaike information criterion (AIC), (2) Bayesian information criterion (BIC), (3) entropy, and (4) the proportion of participants in the smallest class. Lower AIC and BIC values indicate better relative model fit, while higher entropy values indicate clearer separation between profiles. Class size was also considered because very small profiles may be less stable and more difficult to interpret. Across the model solutions, AIC continued to decrease as additional profiles were added. However, BIC worsened at the five-profile solution, and the six-profile solution included a smallest class of only 5% of the sample. The four-profile solution was therefore selected because it retained acceptable entropy (.84), had a smallest class size above 5% of the sample, and produced theoretically meaningful attachment profiles.

Following profile selection, participants were assigned to the profile for which they had the highest posterior probability. Group differences across the attachment profiles were then examined for sleep quality, mood symptoms, and fatigue. Welch's ANOVAs were conducted because the profile groups were not equal in size and the assumption of equal variances could not be assumed. When Welch's ANOVA indicated a significant difference across profiles, Games-Howell post-hoc comparisons were used to identify which specific profiles differed from one another.

3 Results

3.1 Latent Profile analysis

The latent profile analysis indicated that a four-profile solution provided the most interpretable representation of attachment patterns in the sample. The four profiles were labeled as: Dismissive-leaning profile (Profile 1; n = 29, 13.2%), Fearful-avoidant profile (Profile 2; n = 51, 23.2%), Anxious-preoccupied profile (Profile 3; n = 47, 21.4%), and Secure profile (Profile 4; n = 93, 42.3%). The standardized mean scores for the 12 attachment indicators are shown in Figure 1. Scores above zero indicate higher-than-average endorsement of that attachment-related item, while scores below zero indicate lower-than-average endorsement.

Fig.1 shows the standardized scores for the four attachment profiles across the 12 attachment indicators, including both avoidance and anxiety items. The Dismissive-leaning profile (Profile 1) was characterized by slightly elevated scores on some avoidance-related items and generally low scores on anxiety-related items, suggesting a more independent and less reassurance-seeking attachment pattern. The Fearful-avoidant profile (Profile 2) was characterized by consistently above-average scores across both avoidance and anxiety indicators, suggesting that individuals in this group both desired closeness and felt anxious about relationships while also tending to pull away from intimacy. The Anxious-preoccupied profile (Profile 3) generally showed below-average scores on avoidance items but elevated scores on anxiety items, especially reassurance-seeking, suggesting that this group was open to closeness but worried about being loved, cared for, or abandoned. The Secure profile (Profile 4) was characterized by consistently below-average scores across both avoidance and anxiety indicators, suggesting comfort with intimacy, low relationship anxiety, and confidence in partner availability.

3.2 Differences in Demographics Among Attachment Profiles

Demographic differences across the four attachment profiles were examined. Age was compared across profiles using Welch's ANOVA, and categorical de-

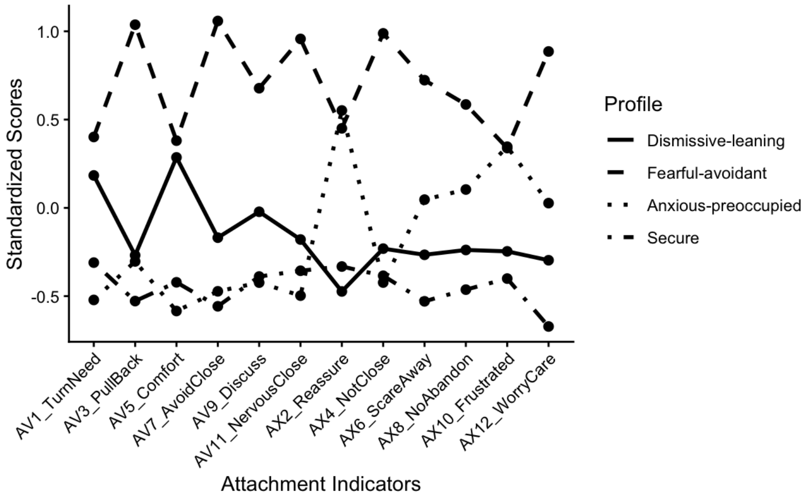

Figure 1: Standardized attachment indicator scores across latent profiles

mographic variables were compared using Pearson's chi-square tests.

Age

The Dismissive-leaning profile was slightly older on average ($M = 35.69$, $SD = 10.28$) than the Fearful-avoidant ($M = 31.69$, $SD = 8.24$), Anxious-preoccupied ($M = 30.53$, $SD = 8.09$), and Secure profiles ($M = 32.58$, $SD = 9.06$). However, this difference was not statistically significant, suggesting that the attachment profiles did not differ meaningfully by age.

Gender

Gender differed significantly across attachment profiles, $\chi^2(3) = 15.20$, $p = .002$. The Anxious-preoccupied profile had the highest proportion of female participants (72.3%), followed by the Fearful-avoidant profile (58.8%) and the Secure profile (51.6%). In contrast, the Dismissive-leaning profile had the lowest proportion of female participants (27.6%).

Ethnicity

Ethnicity also differed significantly across attachment profiles, $\chi^2(12) = 25.99$, $p = .011$. The Dismissive-leaning profile had the highest proportion of White participants (65.5%), while the Anxious-preoccupied profile had the highest proportion of Black, Black British, Caribbean, or African participants (53.2%). The Fearful-avoidant profile also included a relatively high proportion of Black, Black British, Caribbean, or African participants (47.1%). Because some ethnicity categories had small cell counts, this result should be interpreted with caution.

Living with spouse

Living with a spouse or partner did not differ significantly across attachment profiles, $\chi^2(6) = 2.38$, $p = .881$. Across all four profiles, most participants reported living with their spouse or partner, with percentages ranging from 74.5%

to 82.8%.

Relationship duration
Relationship duration also did not differ significantly across attachment profiles, $\chi^2(6) = 9.08$, $p = .169$. In all four profiles, the majority of participants reported being in their relationship for more than three years, suggesting that relationship length was relatively similar across attachment profiles.

3.3 Differences in Sleep and Mood Outcomes Among Attachment Profiles

Results from Welch's ANOVAs indicated that attachment profiles differed across several sleep and mood outcomes, including sleep problems, sleep efficiency, sleep hygiene, anxious mood, depressed mood, and enthusiasm/trouble. Table 2 presents the follow-up Games-Howell post-hoc comparisons.

Sleep Problems
The Fearful-avoidant and Anxious-preoccupied profiles reported the highest levels of sleep problems. Follow-up tests showed that both the Fearful-avoidant and Anxious-preoccupied profiles reported significantly more sleep problems than the Secure profile.

Sleep Efficiency
The Fearful-avoidant profile showed the worst sleep efficiency, whereas the Dismissive-leaning profile showed the best sleep efficiency. Follow-up tests indicated that the Dismissive-leaning profile differed significantly from the Fearful-avoidant profile.

Sleep Hygiene
The Dismissive-leaning profile reported the best sleep hygiene, followed by the Secure profile. In contrast, the Fearful-avoidant and Anxious-preoccupied profiles reported poorer sleep hygiene. Follow-up tests showed that the Dismissive-leaning profile reported significantly better sleep hygiene than both the Fearful-avoidant and Anxious-preoccupied profiles. The Secure profile also reported significantly better sleep hygiene than the Fearful-avoidant and Anxious-preoccupied profiles.

Sleep Quality
The Fearful-avoidant and Anxious-preoccupied profiles reported the poorest overall sleep quality. The Secure profile showed the lowest sleep quality score, suggesting relatively better sleep quality compared with the other profiles.

Sleep Onset
The Anxious-preoccupied profile showed the highest sleep onset score, suggesting the longest time falling asleep. The Fearful-avoidant profile also showed a relatively elevated sleep onset score compared with the Dismissive-leaning and Secure profiles.

Sleep Duration
The Fearful-avoidant profile reported the shortest sleep duration, whereas the

Dismissive-leaning profile reported relatively longer sleep duration. The Secure and Anxious-preoccupied profiles fell between these two groups.

Anxious Mood
The Fearful-avoidant profile reported the highest anxious mood scores, followed by the Anxious-preoccupied profile. The Secure and Dismissive-leaning profiles reported lower anxious mood scores. Follow-up tests showed that the Secure profile differed significantly from both the Fearful-avoidant and Anxious-preoccupied profiles. The Fearful-avoidant profile also reported significantly higher anxious mood than the Dismissive-leaning profile.

Depressed Mood
A similar pattern was observed for depressed mood. The Fearful-avoidant profile reported the highest depressed mood scores, and the Anxious-preoccupied profile also showed elevated depressed mood. Follow-up tests showed that the Secure profile differed significantly from both the Fearful-avoidant and Anxious-preoccupied profiles. The Fearful-avoidant profile also reported significantly higher depressed mood than the Dismissive-leaning profile.

Enthusiasm and Trouble
The Secure profile reported the fewest problems with enthusiasm and daily functioning. Follow-up tests showed that the Secure profile differed significantly from the Fearful-avoidant profile.

Fatigue
The Secure profile reported the lowest level of fatigue, whereas the Fearful-avoidant profile reported the greatest fatigue. The Dismissive-leaning and Anxious-preoccupied profiles fell between these two groups.

Table 2: Sleep, mood, and fatigue outcomes by attachment profile

Variable	Dismissive-leaning	Fearful-avoidant	Anxious-preoccupied	Secure
SLEEP_DUR_SCORE	0.45 (0.15)^{a}	0.88 (0.16)^{a}	0.55 (0.13)^{a}	0.63 (0.09)^{a}
ONSET_SCORE	0.83 (0.18)^{a}	1.20 (0.16)^{a}	1.30 (0.17)^{a}	0.94 (0.11)^{a}
SLP_QUALITY_SCORE	1.10 (0.09)^{a}	1.22 (0.09)^{a}	1.23 (0.09)^{a}	1.01 (0.06)^{a}
Sleep_Problem_score	1.48 (0.14)^{ab}	1.68 (0.10)^{a}	1.70 (0.10)^{a}	1.29 (0.07)^{b}
Enthusiasm_and_trouble_score	1.03 (0.13)^{ab}	1.41 (0.09)^{a}	1.26 (0.12)^{ab}	0.97 (0.06)^{b}
sleep_efficiencySCORE	0.17 (0.10)^{a}	0.80 (0.16)^{b}	0.38 (0.12)^{ab}	0.36 (0.08)^{ab}
SLEEP_HYG_SCORE	48.00 (1.37)^{a}	42.61 (1.05)^{b}	42.65 (0.97)^{b}	46.49 (0.62)^{a}
Anxiousmood	3.55 (0.29)^{ab}	4.92 (0.23)^{c}	4.51 (0.25)^{ac}	3.45 (0.15)^{b}
Depressedmood	3.48 (0.29)^{ab}	4.71 (0.25)^{c}	4.04 (0.24)^{ac}	3.26 (0.15)^{b}
fatigue	3.41 (0.27)^{a}	3.18 (0.17)^{a}	3.26 (0.23)^{a}	3.67 (0.13)^{a}

Note. Classes that do not share a superscript are significantly different based on Games–Howell post-hoc tests, $p < .05$.

4 Discussion

The present study used latent profile analysis to identify distinct patterns of adult attachment based on attachment anxiety and attachment avoidance indicators. A four-profile solution was selected and labeled as Dismissive-leaning, Fearful-avoidant, Anxious-preoccupied, and Secure. These profiles were broadly consistent with the traditional two-dimensional model of adult attachment, in which different combinations of anxiety and avoidance form theoretically meaningful attachment styles, such as secure, dismissing, preoccupied, and fearful attachment (Bartholomew & Horowitz, 1991). Overall, the findings suggest that attachment anxiety and avoidance do not operate only as separate continuous predictors; instead, people show distinct attachment patterns that are meaningfully related to sleep, mood, and daily functioning.

One major finding was that the Fearful-avoidant and Anxious-preoccupied profiles generally showed poorer outcomes than the Secure profile. Both of these profiles were characterized by elevated attachment anxiety, suggesting that anxiety about closeness, rejection, and reassurance may be especially important for sleep and mood outcomes. These two groups reported significantly more sleep problems than the Secure profile, and both also showed poorer sleep hygiene compared with the Dismissive-leaning and Secure profiles. This pattern is consistent with prior research showing that insecure attachment is associated with poorer sleep quality and poorer sleep hygiene (Samii et al., 2026). The present study extends that work by showing that these associations may differ depending on the specific combination of attachment anxiety and avoidance, rather than only the overall level of insecurity.

The Fearful-avoidant profile appeared to be the most vulnerable group overall. This profile had high scores across both anxiety and avoidance indicators and showed worse sleep efficiency, poorer sleep hygiene, higher anxious mood, higher depressed mood, and more problems with enthusiasm and daily functioning. This pattern makes sense theoretically because fearful-avoidant attachment combines two potentially stressful relationship tendencies: wanting closeness but also feeling uncomfortable with or mistrustful of closeness. Prior attachment research suggests that insecure attachment is linked to less effective emotion regulation and stronger stress-related responses, which may contribute to both mood difficulties and sleep disruption (Mikulincer & Shaver, 2019; Pietromonaco & Beck, 2019). In this study, the Fearful-avoidant group may have experienced the combined burden of attachment-related worry and avoidance-based distancing, which could help explain why this profile showed the least adaptive pattern across outcomes.

The Anxious-preoccupied profile also showed elevated difficulties, especially for sleep problems, sleep hygiene, anxious mood, and depressed mood. This group was not strongly avoidant and appeared relatively open to closeness, but it showed high reassurance-seeking and relationship worry. This supports the idea that attachment anxiety may be particularly relevant to pre-sleep arousal and emotional distress. People high in attachment anxiety may be more likely to worry about whether they are loved, whether their partner is available, or whether the relationship is stable. These concerns may make it harder to wind down at night and may also contribute to anxious and depressed mood. This is consistent with broader findings that attachment anxiety and avoidance are both associated with poorer sleep quality in adults, although the mechanisms may differ across attachment patterns (Hu et al., 2024).

The demographic findings also provide important context. Age, living with a spouse or partner, and relationship duration did not differ significantly across profiles, suggesting that the profile differences were not simply explained by these relationship-status variables. However, gender differed significantly across profiles. The Anxious-preoccupied profile had the highest proportion of female participants, while the Dismissive-leaning profile had the lowest proportion of female participants. This does not mean that gender determines attachment style, but it does suggest that anxious-preoccupied attachment was more common among women in this sample. This pattern is broadly consistent with prior research suggesting that women, on average, tend to report somewhat higher attachment anxiety, whereas men tend to report somewhat higher attachment avoidance, although these gender differences are usually small and can vary across samples and cultures (Del Giudice, 2019).

Ethnicity also differed significantly across attachment profiles. The Dismissive-leaning profile had the highest proportion of White participants, whereas the Anxious-preoccupied profile had the highest proportion of Black, Black British, Caribbean, or African participants. This finding should be interpreted with caution because some ethnicity categories had small cell counts. It should not be taken to mean that ethnicity causes attachment profile membership. Rather, it suggests that attachment patterns may be shaped by broader social experiences, family environments, cultural expectations, and structural contexts. Prior research has noted that attachment theory should be applied with attention to cultural context and that attachment-related meanings may vary across cultural groups (Keller, 2013; Sakman & Sümer, 2024). Future research should examine these demographic patterns in larger and more diverse samples.

Several limitations should be noted. First, this study used a cross-sectional design, meaning that all variables were measured at one point in time. Therefore, although the results showed meaningful differences across attachment profiles, they cannot determine whether attachment patterns caused differences in sleep, mood, or fatigue. Future research should use longitudinal data to examine whether attachment profiles remain stable over time and whether they predict later changes in sleep quality, mood symptoms, or fatigue. Second, all measures were based on self-report. As a result, participants' responses may have been influenced by memory errors, social desirability, or individual response styles. Future studies could include more objective sleep measures, such as actigraphy or sleep diaries, to provide a more complete assessment of sleep outcomes. Third, the sample size was relatively modest. Although the sample was large enough to identify meaningful profile patterns, the profile solution should not be treated as final or universal. Replication in larger and more diverse samples would help determine whether the same four-profile structure appears consistently across different populations.

Another limitation is that there is no single definitive statistical rule for selecting the correct number of profiles in latent profile analysis. In this study, profile selection was guided by multiple criteria, including AIC, BIC, entropy, class size, and theoretical interpretability. However, some researcher judgment was still involved. Future work should cross-validate the profile solution in independent samples to test whether the four-profile structure is stable.

Finally, profile assignment in LPA is probabilistic rather than certain. Each participant was assigned to the profile they were most likely to belong to, but they still had some probability of belonging to other profiles. In this study, the average posterior probabilities suggested that the profiles were reasonably well separated, so the most likely profile assignment was used in the main analyses. Future research could use posterior-probability weighting or formal three-step/BCH methods to better account for classification uncertainty when comparing profiles on sleep, mood, and fatigue outcomes.

5 Conclusion

The present study highlights the value of using a person-centered approach to examine adult attachment. Rather than treating attachment anxiety and attachment avoidance only as separate predictors, latent profile analysis identified meaningful subgroups with distinct combinations of attachment-related tendencies. These attachment profiles were associated with different patterns of sleep, mood, fatigue, and daily functioning. In particular, profiles marked by higher attachment anxiety, especially the Fearful-avoidant and Anxious-preoccupied profiles, showed the most difficulty across outcomes. Overall, these findings suggest that attachment-based profiles may be useful for understanding individual differences in sleep hygiene, sleep problems, mood symptoms, and fatigue. These results suggest that future sleep and mental health research may benefit from considering not only whether individuals are insecurely attached, but also how attachment anxiety and avoidance combine within distinct attachment profiles.

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